

UNIVERSAL PLASTIC SHREDDER V2.0

Electrical Control & Power System

Bao Nguyen • Fearghus Tyler • Yaqoub Rabiah • Fox Kang

Portland State University • MCECS • Department of Electrical & Computer Engineering

Sponsor: MME / Ichor Systems (Bridget Lannigan) | Advisors: Prof. Andrew Greenberg, Dr. Jonathan Bird (EE), Robert Paxton (ME)

The Problem

Ichor Systems generates large volumes of thick rigid plastic (PVC, PP, LDPE) during semiconductor equipment manufacturing. Off-the-shelf shredders are either too small for the material (0.25–0.5 in. sheets, 4–12 in. wide) or priced far above the sponsor's per-line budget. The EE team designed and built the electrical control and power system for an industrial-grade plastic shredder; the mechanical structure is delivered by a separate ME team.

Requirements

Must:

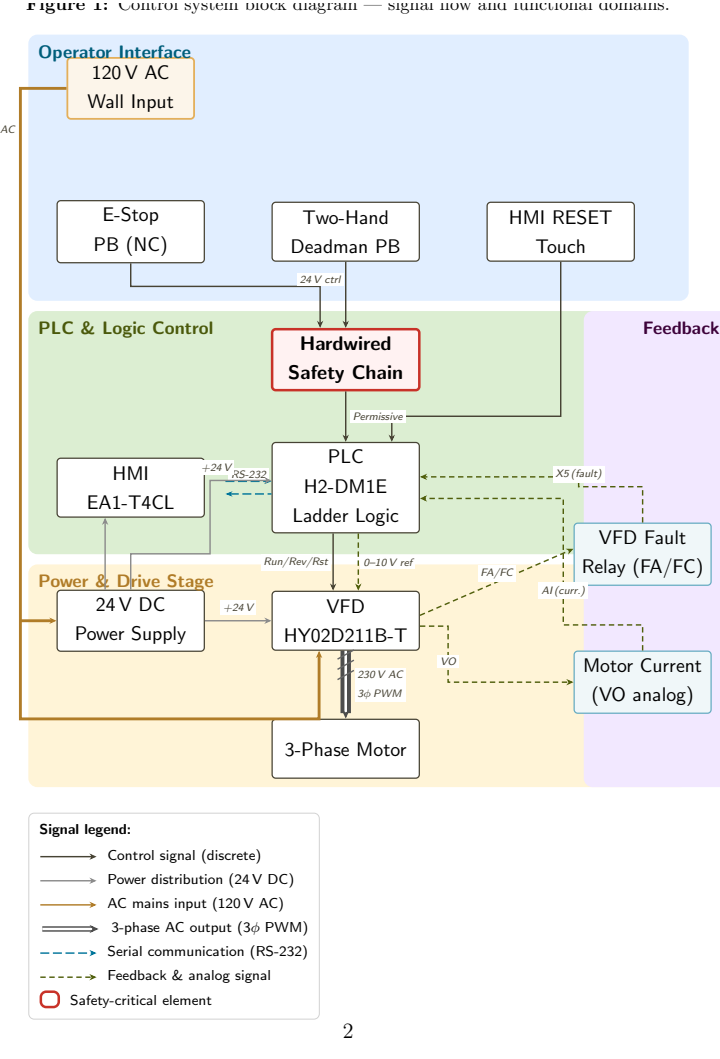
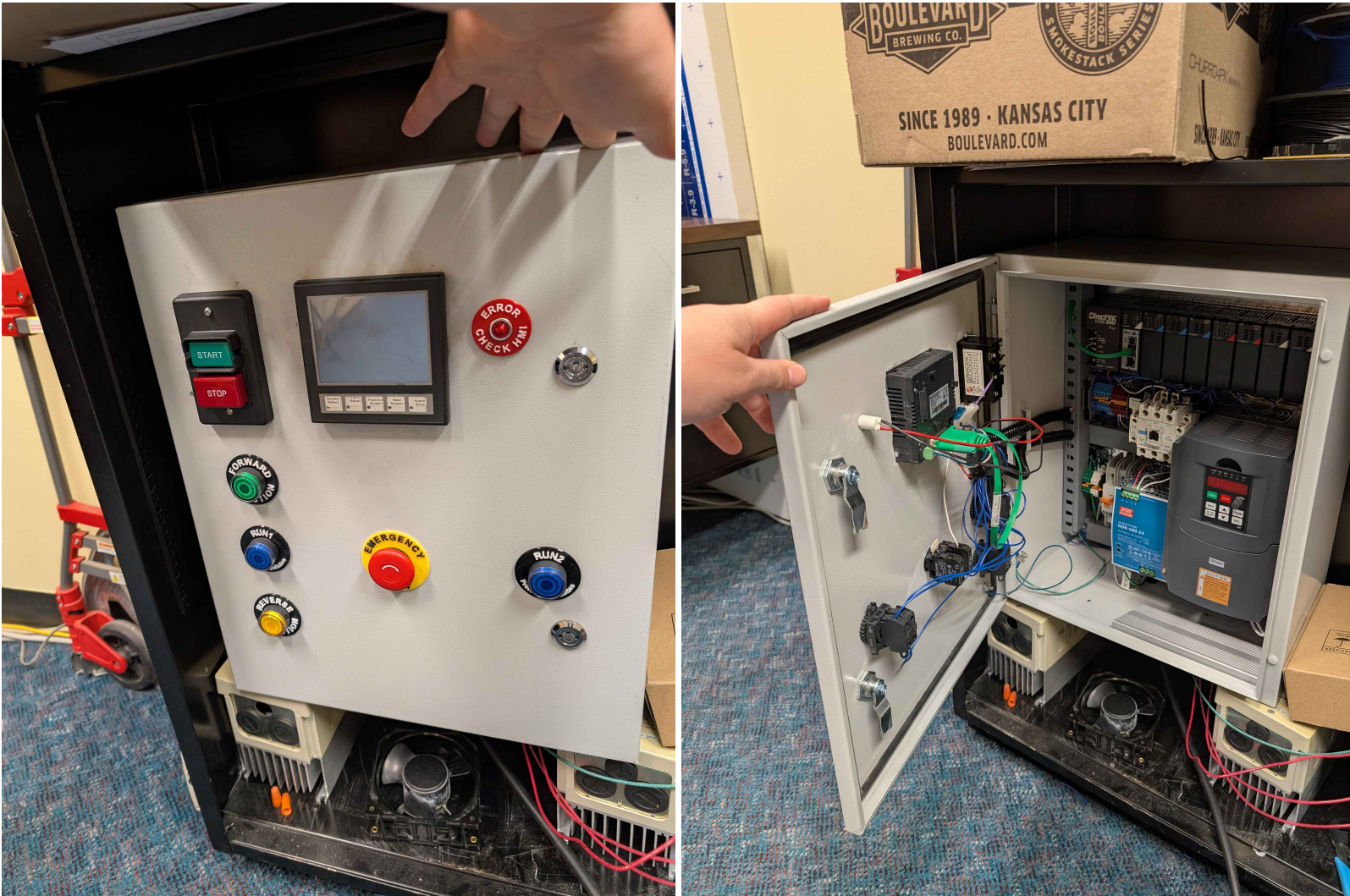
- Shred plastic to ~1 in. pieces.
- Drive a 3–5 HP motor at 70–90 RPM through a controllable VFD.
- Follow NFPA 70 (NEC); include E-stop, overload, and one safety interlock.
- Provide an HMI for operator control and status.
- Stay within the \$1,000 EE-team budget.

Should: status / fault lights, NEMA enclosure, labeled controls.

May: read-only wireless monitoring, data logging, capacitive-touch safety.

System Architecture

The control system is built end-to-end on AutomationDirect industrial hardware – no hobby-grade microcontrollers are in the safety or control path. A PowerTec 71008 disconnect feeds a Mean Well NDR-480-24 24 VDC supply, a Mitsubishi SD-N35 contactor (E-stop hard-wired in series with its coil), and a Huanyang HY02D211B-T VFD. The VFD drives a 2 HP IronHorse MTCP2-002 premium-efficiency TEFC 3-phase induction motor (230 V, 5.93 A FLA, 1735 RPM, 145TC frame, VFD-rated 1.0 SF) over U/W/W via blue, white, and red 4 mm banana sockets.



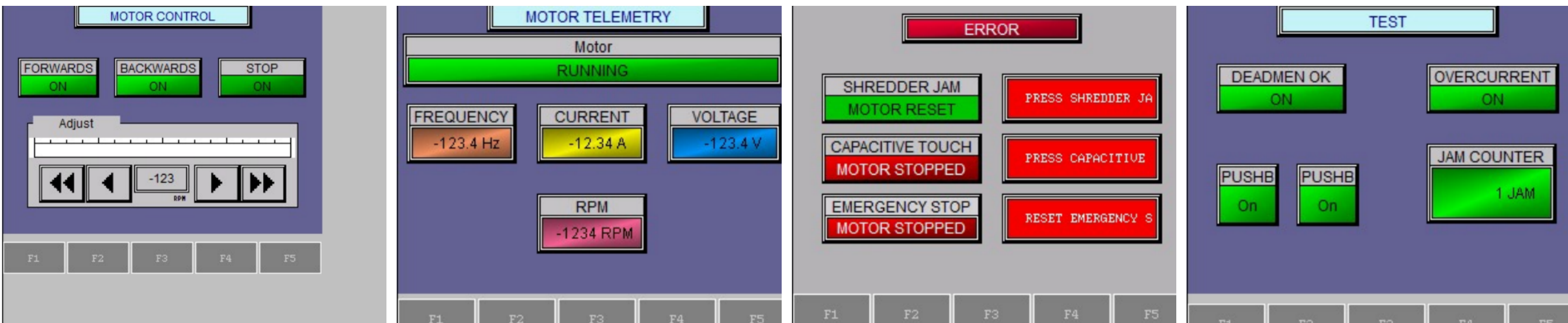
Operator face (left), panel interior (middle), and system block diagram (right): HMI, deadman / fwd / rev / E-stop, DirectLOGIC 205, Huanyang VFD, 24 V supply, SD-N35 contactor, terminal blocks; block diagram shows the signal path from AC in through the safety chain to the VFD and motor.

PLC Control

The PLC is the system controller. A DirectLOGIC 205 D2-09B-1 rack holds the H2-DM1E CPU and five I/O modules: D2-08ND3 (8 DC inputs), F2-08AD-1 (analog current input), F2-04RTD (RTD), D2-08TR (8 relay outputs), and F2-08DA-2 (0–10 V analog output to the VFD). The 31-rung Do-more program implements a deterministic state machine plus a 3-strike jam recovery sequence and a hard-lockout path.

HMI

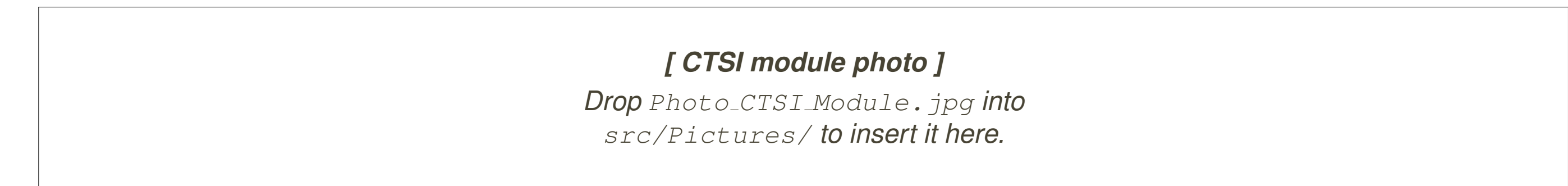
AutomationDirect EA1-T4CL C-more Micro-Graphic 4 in. touch panel running HMI_413.mgp, serial to the H2-DM1E CPU. Four screens hold 27 PLC tags.



Motor Control • Motor Telemetry • Error / Alarm • Test.

Capacitive-Touch Safety (CTSI) Module

A contact-free operator-presence sensor designed and built in-house by the Shredder EE team. An ESP32 reads a capacitive electrode; on a touch event the ESP32 drives an LED through an optocoupler that closes the 24 V PLC input X7. Power is taken from the panel 24 VDC rail through an LM2596 buck converter to the ESP32's 3.3 V rail.



ESP32 + LM2596 buck + optocoupler module wired to PLC X7.

Ladder response (rung 28): touching the electrode latches Y4 (fault LED), C15 (STOP_MOTOR), C21 (CTSI_TRIG), and C22 (CHANGE_SCREEN) – motor stops, fault LED on, HMI jumps to ERROR, all independent of the deadman / E-stop chain.

Status: bench-validated against the PLC; full machine-level verification on the enclosed prototype pending.

Safety & Compliance

- E-stop:** Category 0 stop. Eaton E22B1 NC mushroom in series with the SD-N35 contactor coil. Drops the contactor independently of the PLC.
- Two-hand / deadman:** both pushbuttons must be held to enable the motor (PLC X2 AND X3).
- Overcurrent / jam:** VFD over-torque trip ($PD124 = 150\%$, $PD125 = 3\text{ s}$) on PLC X5; ladder runs the 3-strike unjam routine.
- Enclosure:** Vorksmen SEB001 steel, NEMA Type 12.
- Compliance:** NFPA 70/79, NEMA 250 / ICS / MG 1, ISO 12100, 13849-1 (Cat B–1 / PL b–c), 13850, 13851, OSHA 1910.147.

Results

All EE “Must” requirements met for the items the EE team controls: VFD motor control, NFPA-70 design, E-stop + overload + safety interlock, HMI control & status, and the \$1,000 budget.

Constraint: the ME team did not complete the mechanical shredder build by end of term, so full machine-level integration testing (loaded shredding of plastic samples on the actual frame) was not possible. All EE validation is bench-only against the VFD and motor stand-in; the panel is ready to integrate as soon as the mechanical prototype is delivered.

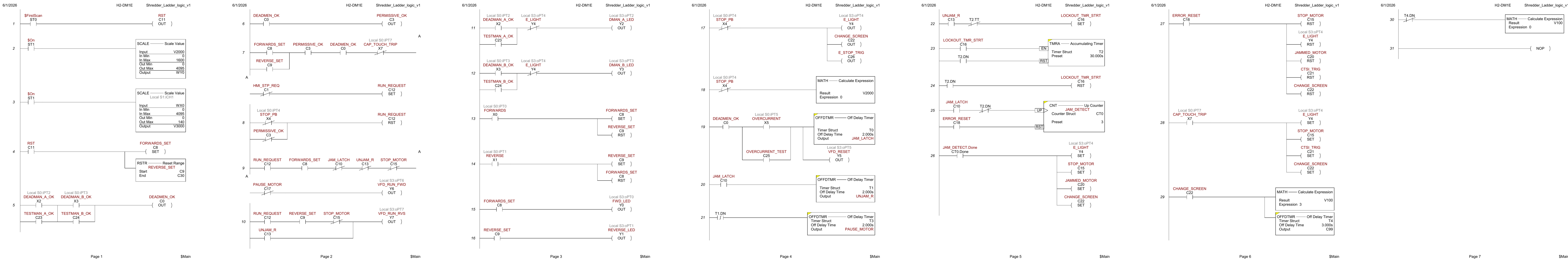
Working:

- Panel built, wired, bench-validated. All 31 rungs program and scan correctly.
- All four HMI screens operate; tag database matches the PLC.
- Speed reference verified: $HMI\ V2000 \rightarrow SCALE \rightarrow F2-08DA-2\ CH1 \rightarrow VFD\ V1$.
- Jam-recovery verified: overcurrent on X5 $\rightarrow 2\text{ s reverse (T1)} \rightarrow$ coast-to-stop buffer (T3).
- Three-revision point list signed off by Fearghus Tyler.

Wired and ready, programming pending: Everything below is fully wired into the panel and the ladder program; what's left is configuration / firmware work, not hardware.

- CTSI module not yet integrated** – the capacitive-touch breakout and its microcontroller were destroyed during bench bring-up. The X7 input, optocoupler interface, and rung-28 ladder response are still in place and ready for the replacement module to drop in.
- Jam-count telemetry not finished** – counter C10 / N0 increments correctly inside the PLC but is not yet surfaced on the HMI for live readout.
- Voltage / current telemetry not working** – plan was to take the VFD's frequency and motor-current analog outputs into the PLC and back-calculate the derived voltage, RPM, and status fields shown on Screen 2. The path $VFD\ VO \rightarrow F2-08AD-1\ CH1 \rightarrow WX0 \rightarrow V3000$ is wired and rung 3 scales it, but VFD analog-output parameters and the PLC scaling still need to be revisited before the live values stabilize.

Complete Ladder Logic — 31 rungs across 7 pages



Pages 1–7: power-up init / deadman • RUN.REQUEST gating / Y6 Y7 outputs • indicator LEDs / direction latches • E-stop / jam detection / unjam stroke • lockout timer / jam counter • fault reset / cap-touch / HMI screen change • screen reset / end of program.

Project By the Numbers

- 31 ladder rungs in production.
- 27 HMI tags across 4 screens.
- 14 component sheets in the point list.
- 3 strikes before the motor locks out.
- 2 s unjam reverse stroke.
- 150% over-torque threshold on the VFD.
- \$969.88 spent against the \$1,000 control-panel budget (\$30.12 under).

Hardware Stack

- PLC:** DirectLOGIC 205 + H2-DM1E.
- HMI:** C-more EA1-T4CL.
- VFD:** Huanyang HY02D211B-T.
- Motor:** IronHorse MTCP2-002 (2 HP, 1735 RPM, 145TC, TEFC, VFD-rated).
- PSU:** Mean Well NDR-480-24.
- Contactor:** Mitsubishi SD-N35.

Future Work

- Production motor selection (3–5 HP) supersedes PD141–PD144.
- Mechanical integration and full shred-load run pending – ME team did not deliver the prototype by end of term.
- Capacitive-touch (CTSI) module on X7: machine-level verification pending.
- Modbus link to the VFD for richer telemetry (deferred – the Modbus module exceeded the \$1,000 control-panel budget, so the analog VO / V1 path was used instead).

Acknowledgments

Thanks to **Bridget Lannigan** (MME / Ichor Systems) for sponsoring the project; to advisors **Prof. Andrew Greenberg**, **Dr. Jonathan Bird**, and **Robert Paxton**; to the **ME capstone team** for the mechanical design and ongoing integration coordination; and to the **PSU Power Engineering Laboratory** for bench access.

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